

The Issue of Gun Violence and the Effectiveness of Gun Laws

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01:790:300:01 Introduction to Political Science Research Methods

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An Introduction to the Issue of Gun Violence:

Gun violence in the United States has become a pressing political and societal issue which threatens public safety, community stability, and American culture. While the plague of gun violence has existed for centuries, it has worsened over the 21st century, with politicians creating division rather than solutions. The argument of more or less gun control restrictions has headlined public policy for years, and despite the 46,726 gun-related deaths in 2023, lawmakers remain divided.

I chose the topic of gun violence as it threatens the everyday lives of Americans, and while the problem seems easily approachable, it remains unsolved. While some states have adopted more and stricter gun policies, others have failed to do so, creating a cultural and political divide within our nation. This topic should be interesting to a general audience as the problem directly affects their personal safety and the future security of their country. In this paper, I will aim to answer the following question: How does a state's gun control laws affect gun violence in the U.S? By studying this question, scholars and policymakers can create evidence-based solutions with real policy implications to address the ongoing gun violence crisis in the United States.

The Relationship Between Gun Laws and Gun Violence:

As the crisis of gun violence within the United States has grown more severe, many researchers and scholars have used statistical analysis to challenge assumptions and find solutions. In *“The Era of Progress on Gun Mortality: State Gun Regulations and Gun Deaths From 1991–2016”*, Kang and Sharkley examine the relationship between state-level gun regulations and gun-related deaths between the years of 1991 and 2016, with the goal of the

paper being to attack the perception that gun violence in America is too difficult to control. The authors focus on which US states implemented stricter gun control laws over the given time period, highlighting how household gun possession rates and gun-related deaths dramatically decreased. The paper concluded that stricter gun policies correlate with a decrease in gun homicides, gun suicides, and overall gun deaths. Specifically, it was found that for each restrictive gun policy enacted, there was a decrease of -0.21 gun deaths per 100,000 residents, demonstrating the relationship between gun regulations and gun-related deaths.

The study “*Firearm legislation and firearm mortality in the USA: a cross-sectional, state-level study*” further proves the findings of the previous article; however, it expands upon which policies are associated with the highest decrease in gun-related death rates. In the study, the authors discuss how different firearm laws, on the state-level, have impacted the rates of firearm suicide, firearm homicide, and overall firearm mortality. The study was conducted following the enactment of firearm policies by many U.S. states in an effort to reduce the firearm mortality rate in the nation. From 2014-2015, the student group constructed a state-level dataset that focuses on gun deaths in each state from 2008-2010 and 25 firearm state laws implemented in 2009. From these 25 new policies, it was found that nine of them correlated with increased firearm mortality rates, nine of them correlated with decreased firearm mortality rates, and seven of them had no statistical significance. However, the three laws that led to the highest decrease in firearm mortality rates were universal background checks for firearm purchase, ammunition background checks, and identification requirements for firearms.

While these laws are correlated with lower gun-related deaths, more aggressive, defensive laws, such as “Stand Your Ground” laws, are often related with higher gun-related death rates. The article, “*State Firearm Laws and Rates of Assault-Related Firearm Death*”,

investigates the relationship between state-level firearm laws and assault-related firearm deaths while also discussing the effects of mandated background checks on firearm mortality rates. The study examines the effects of several gun control laws, including “Stand Your Ground” laws, in all 50 states and Washington, D.C. between 2009 and 2018. The article presents several datasets that showcase how each state compares to each other on the basis of firearm-related assault mortality per 100,000 persons while also showing the effects on assault mortality based on if certain gun policies were absent or present. Through statistical analysis, the study concluded that “Stand Your Ground” laws are associated with higher gun-related assault rates, while background checks on private sales are correlated with a lower rate, as concluded in the previous study. By comparing information across all 50 states, the research in this study directly relates to the dataset and statistical tests that will be presented in this paper.

While each state is responsible for their own gun control laws, the effects of those laws are often not independent on a state-level. In other words, the gun control laws of one state may affect the gun-related deaths of another, demonstrating how state gun control laws can “leak” beyond state borders. The article, “*Association of State-Level Firearm-Related Deaths With Firearm Laws in Neighboring States*”, describes the relationship between state-level firearm laws and firearm-related deaths within states and across borders. The goal of the research design was to account for the limitation of how firearms can travel across borders, helping to demonstrate more accurate state-level analysis between firearm laws and deaths. The study used data from 48 states, excluding Alaska and Hawaii, from 2000-2019, focusing on nine specific firearm laws and their impact. In their findings, they concluded that “Permit Requirement for Firearm Purchase” had the largest impact on in-state and inter-state associations, while also highlighting the gun control laws of one state are not independent of the gun-related deaths of another state.

In response to the issue of gun violence, some organizations have conducted statistical research in an effort to educate the public and create meaningful policy changes. “Everytown for Gun Safety” is a gun violence prevention organization that aims to lobby for gun control laws in an effort to make everyday life safer. In one dataset constructed by the organization, they compared every states’ strength of its gun laws and its rate of gun violence. The strength of each states’ gun policy was determined using a composite score rating of their top 50 gun control laws, and its rate of gun violence measures gun deaths per 100,000 residents. States are ranked using several categories, including “national leaders”, “missing key laws”, and “national failures”. The organization concluded that states that elected officials who implemented stricter gun control laws have faced lower rates of gun violence, and if all states adopted the gun restrictions of national leaders, they predict that there would be 299,000 less gun deaths in the next decade.

Several observations from the database strongly connect to the findings of previously discussed sources. For example, one finding in the dataset concluded that one state’s strict system can’t protect a neighboring state with a weak system, connecting to the findings in *“Association of State-Level Firearm-Related Deaths With Firearm Laws in Neighboring States”*. The organization agrees with the conclusions of the three previous sources regarding which gun policies result in fewer gun deaths by arguing that states should adopt background checks, purchase permitting, and secure gun storage requirements while rejecting “Stand Your Ground” and permitless carry laws. Lastly, several sources directly operationalize gun violence as state gun-related deaths per 100,000 citizens, which will be the measurement used in the following sections of this paper.

Theory and Hypotheses:

Based on the literature, states that adopt more and stricter gun control laws tend to have fewer gun deaths, and therefore lower levels of gun violence. When states adopt stricter gun control laws, they are typically enacting universal background checks, permit requirements, and firearm identification requirements. These policies lead to a reduction in the number of guns and access by high-risk individuals, ultimately resulting in decreases in firearm suicide rates, firearm homicide rates, and overall firearm mortality rates. It is reasonable to assume that states that have a higher number of gun control policies are also stricter about those laws, and the more gun control policies a state enacts, the more likely they are to be adopting policies requiring background checks and specific requirements. Therefore, this theory overcomes the first hurdle of causality as there is a clear and reliable relationship between the independent variable (state gun control laws) and the dependent variable (gun violence). Several complexities exist within this theory, including spillover effects and policy heterogeneity. While there is a strong correlation between a state's gun control laws and their gun-related deaths, a weak gun control state can increase the firearm mortality rate in neighboring states, regardless of those states' gun policy strength. This causes a weaker correlation between the theory's independent variable and dependent variable within an intra-state scope as states' gun control laws and gun-related deaths are not independent of each other. Additionally, gun policies such as "Stand Your Ground" and permitless carry laws lead to an increase in firearm mortality rate, therefore weakening the negative relationship between a state's gun control policies and their gun-related deaths.

My testable hypothesis is as follows: in a comparison of states, those having more gun laws will be more likely to have less gun deaths per 100,000 residents than will those having less

gun laws. In other words, states with a higher number of gun control laws will experience a lower number of gun deaths than states with fewer gun control laws.

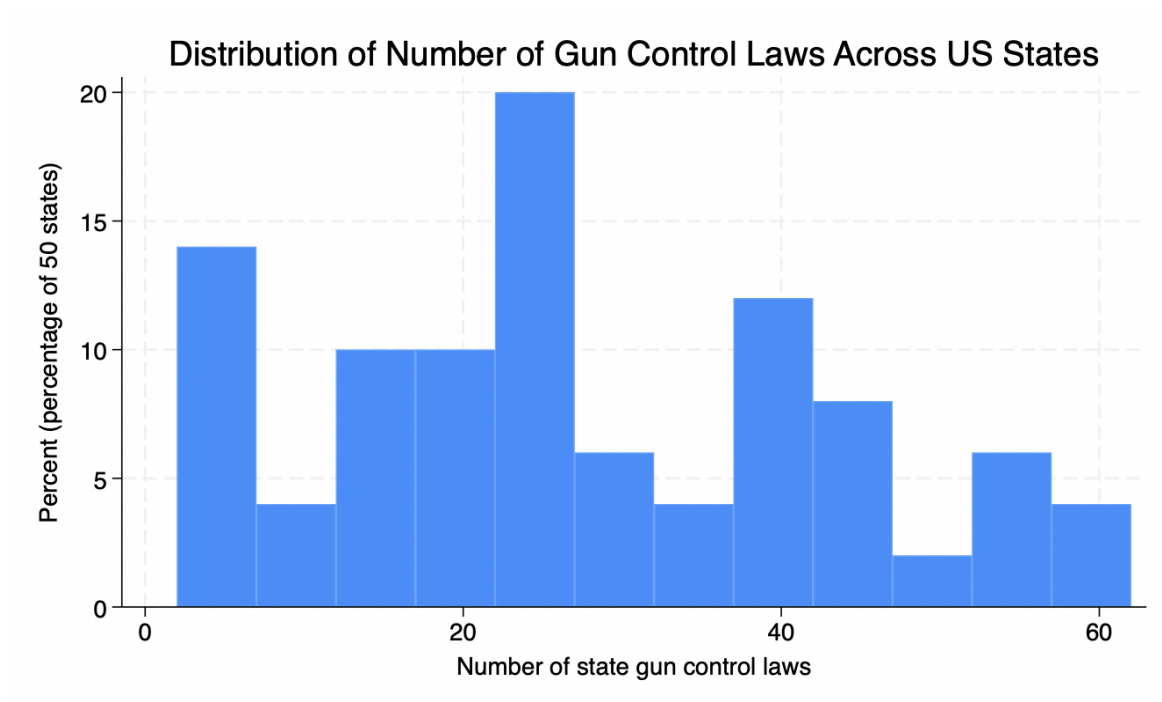
My null hypothesis is there is no relationship between the number of gun control laws within a state and the number of gun deaths in that state.

Research Design:

Before conducting statistical analysis to test my hypothesis, it is necessary to define and describe the variables that will be used in this study. My independent variable of gun laws can be conceptually defined as the extent to which a U.S. state exhibits the characteristic of regulating the ownership, carrying, and use of firearms, and my dependent variable of gun violence can be conceptually defined as the extent to which individuals within society exhibit the characteristic of using firearms to injure or damage themselves or others, regardless of intent. This definition of gun violence encompasses a wide range of gun usage that includes homicides, suicides, or accidental gun-related injuries or death. In order to operationalize these concepts into measures, I will be using Phillip H Pollock's States dataset, which measures a range of variables in a comparison between U.S. states. The independent variable (gunlaws) will be operationalized as the number of gun control laws per state, and the dependent variable (gun_deaths_100k) will be operationalized as the number of gun deaths per 100,000 residents in a state. Both of these variables are continuous as they can be meaningfully ordered and can be equally distanced. For example, gun laws can be meaningfully measured on a numeric scale (1, 2, 3, etc.) and each value is equally distanced from one another. Since gun laws and gun deaths are continuous variables, the mean is the appropriate measure of central tendency for description. The mean of the independent variable, gun laws, in the dataset is 27.18, indicating that on average, a state

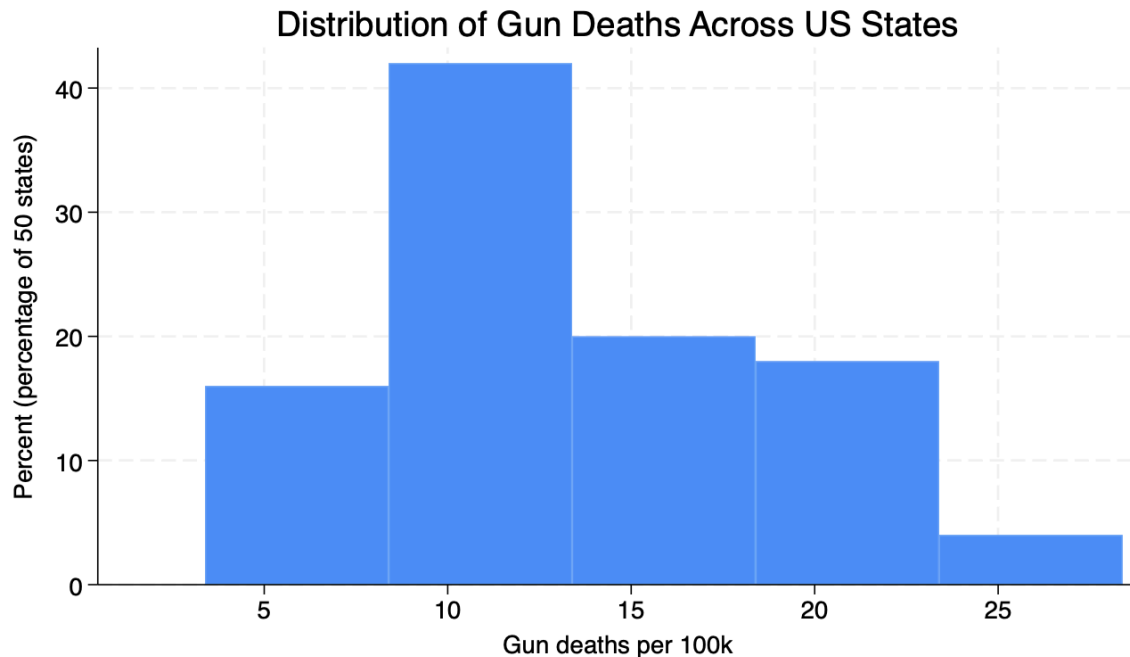
possesses 27.18 gun control laws. For the dependent variable, gun deaths, the mean in the dataset is 13.41, indicating that on average, a state experiences 13.41 gun deaths per its 100,000 residents.

To understand how the values of each variable vary from the mean, it is important to observe the dispersion for both variables. Since the independent variable and the dependent variable are continuous, the appropriate measure of dispersion for both variables is standard deviation.



The standard deviation for variable “gunlaws” in the dataset is 15.93, meaning that on average, states deviate from the mean gun laws by 15.93 gun laws. Given that the variable has a range of 58 gun laws, the standard deviation implies a generally high level of dispersion. As seen in the chart above, the variable exhibits a medium to high level of dispersion. The values are clustered throughout the variable, with five categories (ex. 18-22 gun laws) garnering 10% or

more of the variable. No category reaches above 20%, meaning there is no value that dominates the variable.



The standard deviation for variable “gun_deaths_100k” is 5.46, indicating that on average, states deviate from the mean gun death rate by 5.46 per 100,000 residents. Based on the shape and spread seen in the graph above, the variable exhibits a medium level of dispersion. For example, over 40% of states have about 9-13 gun deaths per 100,000 people, and states with 4-8, 14-18, and 19-23 gun deaths per 100,000 all consume about 15-20% of the sample. This suggests that the dispersion is neither heavily skewed in any direction nor evenly dispersed across the sample.

One variable that could influence the dependent variable of gun violence is the poverty rate within a state. Disadvantaged communities face a higher level of gun violence as a result of their lack of resources, seen in poor legal enforcement, inadequate education, and financial pressures. This results in an increase in crime rate, which is heavily carried out by the use of

firearms. Individuals within communities of high crime rates often feel the need for self-protection, which results in an increase in firearms and usage. Since higher poverty rates are associated with higher gun violence and possibly an increase in the need to pass more gun control laws, the negative correlation between gun violence and gun laws should strengthen when controlling for poverty rate, making the relationship more apparent. In the following statistical analysis, poverty rate will be operationalized through the variable “poverty_rate”, which describes the percentage of people in poverty within each state. Since the values of the variable can be meaningfully ordered and are equally distanced, it is a continuous variable, meaning that the appropriate measures of central tendency and dispersion are mean and standard deviation respectfully. The mean of the control variable in the dataset is 12.14, indicating that the average poverty rate per state is 12.14%, and the standard deviation of the variable is 2.7, meaning that on average, states differ from the mean poverty rate by 2.7 percentage points.

Statistical Analysis & Discussion of Results:

Based on the nature of the independent variable, dependent variable, and control variable, I will first be running a bivariate regression test between my independent and dependent variable followed by a multivariate regression test including my control variable. These statistical tests are appropriate for the research tests as all variables used are continuous. A regression test shows the direction and strength of the variables’ relationship, producing a beta coefficient that estimates the effect between two variables. The regression coefficient helps predict change in the dependent variable for a one-unit change in the independent variable, strengthening the case for a given hypothesis or theory.

<code>. reg gun_deaths_100k gunlaws</code>						
Source	SS	df	MS	Number of obs	=	50
Model	596.558382	1	596.558382	F(1, 48)	=	33.16
Residual	863.454533	48	17.9886361	Prob > F	=	0.0000
				R-squared	=	0.4086
				Adj R-squared	=	0.3963
Total	1460.01292	49	29.7961819	Root MSE	=	4.2413
gun_dea~100k	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
gunlaws	-.2190795	.038043	-5.76	0.000	-.29557	-.142589
_cons	19.36818	1.195385	16.20	0.000	16.9647	21.77166

The bivariate regression test produced a p-value of 0.000, so there is less than a 5% probability that the relationship between a state's number of gun laws and gun deaths is due to random chance. Since the p-value is less than 5%, we reject the null hypothesis of no relationship between the number of state gun laws and state gun deaths per 100,000 residents; therefore, we conclude there is a statistically significant relationship between the two variables. Additionally, a one-unit increase in the number of gun laws per state is associated with a -0.22-unit decrease in state gun deaths per 100,000 people, indicating that gun laws have a negative effect on gun deaths. The 95% confidence interval associated with this coefficient is [-0.30, -0.14], meaning that we are 95% confident that the true effect of a one-unit increase in the number of gun laws is between 0.14 and 0.30 fewer gun deaths per 100,000 residents. The regression test produced an R-squared value of 0.41, indicating that about 41% of the variability in gun deaths per 100,000 residents can be explained by a state's number of gun laws.

Given the results, the bivariate regression between gun laws and gun deaths supports my alternative hypothesis that states with a higher number of gun laws are more likely to have fewer gun deaths per 100,000 residents than states with fewer gun laws. This is due to the p-value

being well under 0.05, indicating a statistically significant relationship between a state's number of gun laws and their number of gun-related deaths. After running a correlation matrix, the correlation coefficient was -0.64, indicating a moderately strong and negative correlation. As predicted in my theory, the correlation between gun laws and gun deaths is slightly limited in strength. This is likely due to the fact that some gun control policies lead to an increase in gun deaths, such as "Stand Your Ground" laws, and states experience a higher firearm mortality rate due to the gun policy strength of a neighboring state.

<code>. reg gun_deaths_100k gunlaws poverty_rate</code>						
Source	SS	df	MS	Number of obs	=	50
Model	918.674661	2	459.33733	F(2, 47)	=	39.88
Residual	541.338254	47	11.5178352	Prob > F	=	0.0000
Total	1460.01292	49	29.7961819	R-squared	=	0.6292
				Adj R-squared	=	0.6134
				Root MSE	=	3.3938
gun_dea~100k	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
gunlaws	-.1744491	.0315893	-5.52	0.000	-.2379986	-.1108996
poverty_rate	.9860154	.1864502	5.29	0.000	.610926	1.361105
_cons	6.180956	2.670793	2.31	0.025	.8080132	11.5539

The multivariate regression test measures the same independent variable and dependent variable; however, it includes poverty rate as a control variable. Similar to the previous test, since the p-value is less than 0.05, there is less than a 5% probability that the observed relationship between the independent and dependent variables are due to random chance even after controlling for poverty rate. Therefore, we can conclude that the relationship between the number of gun laws and the number of gun-related deaths is still statistically significant after

accounting for poverty rate. Furthermore, a one-unit increase in the number of state gun laws is associated with a -0.17 -unit decrease in state gun deaths per 100,000 residents, indicating that gun laws still have a negative effect on gun deaths while controlling for poverty rate. Lastly, the model produced a R-squared statistic of 0.63, indicating that a state's number of gun laws and poverty rate can explain 63% of the variability in gun deaths per 100,000 residents.

When including the control variable of poverty rate, the results indicate a slight decrease in the strength of the relationship between the number of gun laws and the number of gun-related deaths, evident in a smaller coefficient (-0.17). This suggests that poverty rate is also strongly associated with gun violence, as the model's R-squared value increases from 0.41 in the original bivariate regression test to 0.63 after controlling for poverty rate, meaning the two variables explain a large variation in gun deaths. However, the relationship between gun laws and gun deaths remains statistically significant, upholding my theory and demonstrating that while poverty independently increases gun deaths, it does not eliminate the effects of gun laws.

Conclusion:

The overall findings of the paper, between the literature review and statistical analysis, conclude that there is a strong relationship between a state's gun control policy and gun violence. In order to better understand my theory, it is necessary to examine other possible causes of gun violence. As seen in the multivariate regression test, poverty rate was heavily associated with gun deaths, and it's highly likely that several other variables, such as mental health, are related as well. By understanding how the issue of gun violence in the United States is caused, it will become easier to find evidence-based solutions. As previously mentioned, two main limitations exist within my theory: spillover effects and policy heterogeneity. In order to address gun deaths

at a state level, it is important to approach the situation from a national level as state gun control laws are often not independent of one another, and it necessary to enact control laws that lead to a reduction in the number of guns and access by high-risk individuals, such as universal background checks, permit requirements, and firearm identification requirements.

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